

REVIEW ARTICLE

The Impact of ChatGPT on Students' Academic Achievement: A Meta-Analysis

Zhiwei Liu  | Haode Zuo  | Yongjing Lu 

College of Mathematical Science, Yangzhou University, Yangzhou, China

Correspondence: Haode Zuo (yzzxzd@yzu.edu.cn)

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ABSTRACT

Background: ChatGPT, a generative artificial intelligence (GenAI) chatbot, has gained significant traction as a tool for supporting students learning. Despite its growing popularity, there is still no academic consensus on its effectiveness in enhancing students' academic achievement.

Objectives: This study aims to explore the effect of ChatGPT on students' academic achievement through a meta-analysis. It seeks to identify the overall effect size and examine variations based on moderators such as educational level, discipline, intervention duration, sample size, knowledge type, instructional model, role-setting, learning approach, and generated content.

Methods: A meta-analysis was conducted on 37 studies (comprising 37 effect sizes) published between 2022 and 2025. The studies were analysed to calculate the overall effect size (Hedges' g) and to explore subgroup differences.

Results and Conclusion: The findings reveal that ChatGPT has a moderately positive impact on students' academic achievement, with an overall effect size of $g = 0.577$ (95% CI [0.395, 0.759], $p < 0.001$). Further analysis of moderating variables indicates that no significant differences are observed across educational levels, role-setting, or learning approaches. A greater effect is observed in the social sciences compared to other disciplines; an intervention duration of 5–10 weeks has a larger impact on academic achievement compared to other durations; sample sizes ranging from 21 to 40 participants exhibit a larger impact on academic achievement than other sample sizes; ChatGPT is more effective in supporting the learning of declarative knowledge compared to procedural knowledge; the combination of traditional classrooms with ChatGPT is more effective than using ChatGPT in a flipped classroom; compared to generating code, using ChatGPT to generate text has better academic achievement.

1 | Introduction

ChatGPT, one of the most prominent applications of GenAI, has rapidly become a transformative tool in educational environments (Yan et al. 2024). As a large language model developed by OpenAI, ChatGPT is designed to generate human-like text and support personalised learning (Chiu 2024). Due to its ease of integration into digital learning platforms, ChatGPT

has been widely applied in education (Zhang and Tur 2024). Educators and policymakers in countries such as the United Kingdom, the United States, and China have shown growing interest in using ChatGPT to enhance students' academic achievement (Jin et al. 2025). Despite ChatGPT's growing popularity in education, there is no consensus within the academic community about its effectiveness in enhancing students' academic achievement. Several empirical studies

Summary

- What is already known about this topic?
 - Generative AI (GenAI) has been studied in educational contexts, demonstrating potential benefits in creativity, engagement, and motivation, but prior research often focused on GenAI tools in general.
 - ChatGPT, developed by OpenAI, is among the first and most widely used GenAI systems, marking many users' initial exposure to such technologies.
- What this paper adds?
 - This study focuses specifically on ChatGPT, confirming a moderately positive overall effect on students' academic achievement through meta-analysis.
 - Key moderators include discipline, intervention duration, sample size, knowledge types, instructional model, and generated content, each influencing the effectiveness of ChatGPT differently.
- Implications for practice and/or policy?
 - Educators should explore the integration of ChatGPT with virtual tools, which can compensate for ChatGPT's limitations in hands-on experimentation.
 - Educators should develop adaptive models that balance novelty and over-reliance to maintain learner interest and prevent burnout.
 - Moderately sized groups (21–40 participants) optimise ChatGPT's impact by balancing interactivity and feedback.
 - Educators should explore hybrid learning models that integrate ChatGPT with diverse feedback mechanisms to enhance the acquisition of different types of knowledge.
 - Educators should explore a hybrid instructional model that combines flipped classroom elements with traditional classroom settings and ChatGPT support.
 - Text-based tasks can be enhanced through ChatGPT's drafting and feedback capabilities, while code-based tasks benefit from its logic and error correction tools.

have found that ChatGPT contributes positively to student learning by enhancing their learning perception, problem-solving skills, and critical thinking (Mahapatra 2024; Urban et al. 2024; Essel et al. 2024). For example, Rashed Ibraheem Almohesh (2024) suggested that ChatGPT significantly improved students' perceptions of learning autonomy. Similarly, Yilmaz and Yilmaz (2023) also showed that students using ChatGPT in computational assignments demonstrated higher reflective thinking.

However, some studies have reported contrasting results. Dasari et al. (2024) argued that ChatGPT did not significantly improve academic achievement and may even hinder independent cognitive development. Similarly, Johnson et al. (2024) revealed that students who used ChatGPT to write programmes had weaker programming skills than those who did not. Currently, while several reviews have discussed the effect of ChatGPT on students' academic achievement (Heung and Chiu 2025; Deng et al. 2024), the overall effect remains unclear and there

is no consensus on the key moderators influencing this effect. Therefore, this study aims to comprehensively evaluate the impact of ChatGPT on students' academic achievement using a meta-analysis approach.

Meta-analysis offers a systematic and quantitative method for assessing the effectiveness of ChatGPT by integrating effect sizes from multiple empirical studies (Borenstein et al. 2021). This method enables researchers to more accurately estimate the impact of interventions while considering moderator variables. This study not only explores the overall effect of ChatGPT, but also examines potential moderator variables, including educational level, discipline, and others. The findings are expected to provide an empirical foundation for the effective integration of ChatGPT into educational practice.

2 | Literature Review

2.1 | ChatGPT

ChatGPT is a GenAI tool based on the generative pre-trained transformer model (Wong et al. 2024). By using a powerful machine learning system, it can generate contextually coherent and grammatically correct text based on user input (Lo et al. 2024). With the continuous development of ChatGPT's capabilities, it has increasingly been regarded as a tool for offering feedback to students learning (Chiu 2024). Particularly, ChatGPT can perform critical analysis based on real-time input from students, thereby improving learning strategies (Li et al. 2024). This dynamic assessment of learning processes underscores its potential to enhance students' academic achievement. Specifically, ChatGPT can provide instant responses and refine information in different styles, helping students better understand knowledge (Pan et al. 2025). It can also assist students in writing essays by offering grammar and spelling suggestions (Chan et al. 2024). In terms of course design, ChatGPT assists educators in improving their lesson plans (Karaman 2024). By generating course-related discussion topics, ChatGPT can inspire students to engage in deeper discussions (Chang et al. 2024). Most importantly, the application of ChatGPT in education has extended beyond the boundaries of traditional classrooms (Zhou and Kim 2024). For instance, it can function as an intelligent teaching assistant, providing immediate feedback on online learning platforms (Hsu 2024). Overall, ChatGPT not only makes education more flexible and efficient but also drives innovation in educational philosophy and teaching practices (Kucuk 2024; Alneyadi and Wardat 2023).

2.2 | Academic Achievement

To more comprehensively assess the impact of ChatGPT on student learning, this study employs the term academic achievement to represent the learning outcomes and effects achieved by students after using ChatGPT. In education, academic achievement is a multidimensional concept, which often refers to the quality and level of learners' academic activities. Academic achievement can be understood as the mastery of specific knowledge or skills (Çalik and Kurt 2024; Çalik et al. 2024) or as a positive shift in affect and attitude

(Camacho-Morles et al. 2021). It encompasses not only traditional test scores and grades, but also many aspects of students' cognitive development, problem-solving skills, critical thinking skills, and learning motivation. In this meta-analysis, academic achievement serves as a comprehensive criterion for evaluating the overall learning outcomes of students after using ChatGPT. The assessment spans multiple dimensions and can be quantified through measures such as exam scores, paper achievement, and the quality of completed assignments (Deng et al. 2024). Therefore, the study regards academic achievement as a combination of students' learning effects and outcomes (Sun and Zhou 2024, 3), including cognitive-level understanding of knowledge and skills (e.g., creative thinking) and non-cognitive level attitudinal achievement (e.g., interests, motivation).

2.3 | The Effect of ChatGPT on Academic Achievement

Recent literatures on ChatGPT have shown its positive effect on students' academic achievement across different learning areas (Deng et al. 2024; Heung and Chiu 2025), including writing skills (Chan et al. 2024), essay feedback (Mahapatra 2024), musical skills (Zhou and Kim 2024), and medical knowledge (Roganović 2024). For instance, Maghamil and Sieras (2024) found that ChatGPT significantly improved students' writing quality in terms of organisation, grammar, and content. Similarly, Yıldız (2023) found that the use of ChatGPT in language learning environments positively influenced students' learning motivation.

However, some studies have found that ChatGPT has no significant effect on students' academic achievement (Ali et al. 2024; Lo et al. 2024; Sun et al. 2024). For example, Bašić et al. (2023) argued that students using ChatGPT for writing may rely on their prior experiences, potentially leading to poorer writing performance. Ba et al. (2024) also revealed that ChatGPT-assisted instruction did not influence the acquisition of medical theoretical knowledge. Therefore, the impact of ChatGPT on students' academic achievement has not been widely confirmed. Given these conflicting findings, we hypothesise that:

H1. *ChatGPT has a significant positive impact on students' academic achievement.*

2.4 | Educational Levels

Educational levels refer to the stages of education where ChatGPT is applied, which can be categorised as K–12 and college education according to Deng et al.'s (2024) study. Prior research suggests that students at different educational levels vary in digital literacy, cognitive abilities, and learning strategies (Alemdag 2025; Zhu et al. 2025). Specifically, college students often possess stronger technical proficiency and self-directed learning strategies, enabling them to derive greater benefit from ChatGPT-assisted learning (Song and Song 2023). In contrast, the limited cognitive scope and unfamiliarity with complex interfaces among K–12 students may hinder their effective use of

ChatGPT, thereby limiting its overall effectiveness (Chen and Chang 2024). Therefore, we hypothesize that:

H2. *Educational level is a significant moderator in the effect of ChatGPT on students' academic achievement.*

2.5 | Disciplines

Disciplines refer to the types of academic subjects taught with ChatGPT, which can be categorised into information engineering, social sciences, and natural sciences (Sun and Zhou 2024). Existing studies have shown that the characteristics and objectives of different disciplines, as well as the application of ChatGPT within them, may influence students' academic achievement (Zhang and Tur 2024). For example, Zhu et al. (2023) emphasised that the use of ChatGPT should be optimised based on students' disciplinary background and the subject matter, rather than applied uniformly across all disciplines. In natural sciences and information engineering, ChatGPT is often used for experimental simulation and data analysis, particularly in data-intensive analyses and complex computations (Kosar et al. 2024; Küchemann et al. 2023). In social sciences, ChatGPT is often used for text generation and language learning (Mugableh 2024). Therefore, discipline may be an important factor for ChatGPT to improve students' academic achievement. We hypothesise that:

H3. *Discipline is a significant moderator in the impact of ChatGPT on students' academic achievement.*

2.6 | Intervention Durations

According to the study by Sun and Zhou (2024), intervention duration is measured in weeks and categorised as follows: > 15, 11–15, 6–10, 1–5, and < 1 week. Existing literature has shown that AI chatbots have a significant effect on students' academic achievement across different intervention durations (Wang and Fan 2025; Wu and Yu 2024). For instance, Urban et al. (2024) found that students using ChatGPT exhibited greater interest in solving creative problems during a short intervention. Gan et al. (2024) also revealed that students using ChatGPT had a better understanding of medical knowledge after a short-term intervention. However, Avello-Martínez et al. (2024) pointed out that using ChatGPT did not significantly improve narrative writing skills in the short term. Svendsen et al. (2024) also found that ChatGPT had no statistically significant effect during the short-term intervention period. Given the inconsistencies in existing findings, we hypothesise that:

H4. *Intervention duration is a significant moderator in the impact of ChatGPT on students' academic achievement.*

2.7 | Sample Size

The sample size is the number of learners who participated in the ChatGPT intervention activities and can be categorised as 0–20, 21–40, and > 40 (Sun and Zhou 2024). Sample size has a significant impact on students' academic achievement

(Breton 2014). Smaller sample sizes can enhance personalised interaction and attention, thereby facilitating a deeper understanding of learning content for students (Ji et al. 2023). In contrast, larger sample sizes provide broader perspectives, enabling students to benefit from others' viewpoints and experiences (Hobert et al. 2023). In other words, in large classes, ChatGPT primarily serves as an auxiliary tool, addressing the breadth of learning; in smaller classes, it plays a more prominent role in supporting deep learning and fostering more personalised thinking (Bucea-Manea-Țoniș et al. 2022). Therefore, to explore the effects of ChatGPT on different sample sizes, we propose:

H5. *Sample size is a significant moderator in the effect of ChatGPT on students' academic achievement.*

2.8 | Knowledge Types

Knowledge type refers to the categories of knowledge acquired through using ChatGPT. Knowledge type has a significant effect on students' academic achievement and can be categorised into declarative and procedural knowledge (Wang et al. 2024). Specifically, declarative knowledge primarily involves theories, concepts, and the relationships between them, with a relatively abstract learning process (Hsu 2024). In contrast, procedural knowledge focuses on how to perform tasks or procedures (Karaman 2024). In this domain, ChatGPT has the potential to foster creative thinking and improve cognitive abilities (Salem et al. 2025). Therefore, to explore the effect of ChatGPT on academic achievement in different types of knowledge, we propose:

H6. *Knowledge type is a significant moderator in the impact of ChatGPT on students' academic achievement.*

2.9 | Instructional Models

Instructional models refer to the teaching approaches used to implement ChatGPT-based learning and can be categorised into traditional and flipped classrooms (Strelan et al. 2020). Existing research indicates that flipped classrooms can significantly impact academic achievement by shifting decision-making power of learning from teachers to students (Yilmaz and Yilmaz 2023). For example, Huesca et al. (2024) highlighted that integrating ChatGPT into the flipped classroom facilitated a more interactive and personalised approach to teaching. Li (2023) also found that the use of ChatGPT in flipped classrooms effectively enhances students' cognitive development. Furthermore, Essel et al. (2024) found that integrating ChatGPT into the flipped classroom supported the development of students' critical and creative thinking. Based on these results, we propose:

H7. *Instructional model is a significant moderator in the impact of ChatGPT on students' academic achievement.*

2.10 | Role Setting

Role setting refers to assigning specific roles and conditions when interacting with ChatGPT (Kong et al. 2023). ChatGPT's

outputs are influenced by the content of the prompts, highlighting the importance of students effectively using prompts in various contexts (Escalante et al. 2023). Chang et al. (2024) indicated that setting role prompts could significantly enhance the effectiveness of ChatGPT. Rakap (2024) found that setting precise prompts enabled ChatGPT to provide more accurate feedback. Li et al. (2024) also suggested that setting ChatGPT with a specific role, such as I am an undergraduate student, when inquiring about research topics led to more useful responses. Based on these findings, we propose:

H8. *Role setting is a significant moderator in the effect of ChatGPT on students' academic achievement.*

2.11 | Learning Approach

Individual learning and collaborative learning are the two primary approaches in a GenAI-driven learning environment (Sun and Zhou 2024). Group collaboration may be an effective way to promote human-ChatGPT collaborative learning (Zhu et al. 2023). ChatGPT can facilitate knowledge sharing and foster team spirit in collaborative learning (Ji et al. 2023; Sánchez-Ruiz et al. 2023). With personalised feedback from ChatGPT, students can independently build knowledge and improve skills based on their individual needs in self-directed learning (Pan et al. 2025). Based on these, we assume:

H9. *Learning approach is a significant moderator in the impact of ChatGPT on students' academic achievement.*

2.12 | Generate Contents

Existing literature suggests that students exhibit varying levels of familiarity when using GenAI to generate different types of content, which may lead to differences in academic achievement (Sun and Zhou 2024). Usually, text generation is associated with text processing, while code generation requires logical thinking (Schnotz 2002). Text processing tools are well-developed, and students generally possess a high level of proficiency in using them (Darmawansah et al. 2025). In contrast, code generation requires strong algorithmic thinking and logical analysis, potentially imposing a greater cognitive load (Kosar et al. 2024). Therefore, exploring the types of content generated by ChatGPT is crucial for academic achievement. We propose:

H10. *Generation content is a significant moderator in the effect of ChatGPT on students' academic achievement.*

2.13 | Previous Meta-Analysis of ChatGPT in Education

Although several meta-analyses have investigated the effects of ChatGPT on students' learning outcomes, most of them seem to focus on medical education (Liu et al. 2024; Jin et al. 2024; Levin et al. 2024). This represents a significant research gap, as the impact of ChatGPT on student learning in different academic fields remains underexplored. Heung and Chiu's (2025) meta-analysis, which synthesised findings from

17 empirical studies, revealed that ChatGPT is an effective tool for enhancing student learning engagement. However, their analysis did not comprehensively discuss students' academic achievement, and treating engagement as the sole outcome variable was insufficient for a full understanding of ChatGPT's educational effect. It is also worth noting that the number of studies included in the relevant meta-analyses is limited. For example, Rabiou and Nuhu's (2024) meta-analysis of 12 early studies conducted between 2022 and 2023 revealed the positive effect of ChatGPT on student learning process. Wang and Fan's (2025) meta-analysis found that ChatGPT improves higher-order thinking, learning perception and performance. However, specific moderator variables (e.g., sample size, instructional model, etc.) were not considered in this meta-analysis. Furthermore, ChatGPT itself does not directly affect students' academic achievement. The way it is applied is crucial to its effectiveness. Therefore, both the experimental and control groups should use the same learning or teaching methods to ensure comparability. However, in one of the sample studies included by Wang and Fan (2025), the experimental group used a ChatGPT-assisted problem-based learning approach, while the control group did not use a problem-based learning approach (e.g., Hui et al. 2025). This omission may reduce the rigour of the study. Moreover, Deng et al.'s (2024) study has greatly advanced our understanding of the effect of ChatGPT on student learning. However, there is a limitation in the selected studies. These studies should reflect students' actual academic achievement after using ChatGPT as a learning aid. Simply using ChatGPT as a general tool without any learning process involved cannot fully demonstrate its true effect. For example, the study conducted by Ahmed Moneus and Al-Wasy (2024) focused solely on using ChatGPT for translation tasks, which did not provide evidence of its effect on student learning.

In summary, while existing reviews provide evidence of ChatGPT's effect on students' academic achievement, they are constrained by narrow disciplinary focuses, a single outcome variable, limited sample sizes, and insufficient examination of moderator variables. Moreover, some studies do not position ChatGPT as an aid tool in the learning process and fail to effectively control confounding variables that influence the learning process, such as differences in teaching methods between the two groups. These limitations underscore the need for a more comprehensive synthesis of empirical evidence, with a focus on academic achievement and systematic consideration of moderators.

Therefore, using a meta-analysis method, this study aims to examine the overall effectiveness of ChatGPT in enhancing students' academic achievement and address the methodological limitations of previous studies by including a broader range of empirical studies across various educational contexts. Particularly, this study examines potential moderators, including educational level, discipline, intervention duration, sample size, learning approach, generated content, knowledge type, role setting, and instructional model. Through this analysis, the study seeks to provide a more comprehensive understanding of the impact of ChatGPT on students' academic achievement. Accordingly, the study addresses the following research questions:

RQ1. *Is ChatGPT effective in enhancing students' academic achievement?*

RQ2. *What are the significant differences in the impact of ChatGPT on students' academic achievement based on different moderators?*

3 | Method

3.1 | Data Sources and Searching Strategy

To ensure comprehensive inclusion of relevant literature, this study used databases such as Web of Science, ProQuest, ScienceDirect, EBSCOhost, PsycINFO, IEEE Xplore, and Scopus, supplemented by Google Scholar for additional search. A citation tracking method was used for snowball sampling of the collected literature, including influential educational technology journals such as *Computers & Education*. Given that ChatGPT was released on November 30, 2022, only studies published after this date were included in this meta-analysis, with the search conducted in May 2025.

Additionally, the search terms consist of two groups: the first group includes terms associated with ChatGPT, including ChatGPT*, chat generative pre-trained transformer, chatbots, GenAI, generative artificial intelligence, voice generation, text generation, code generation, video generation, and image generation; the second group includes terms related to the research context, such as education, student, and learner. The study initially connected terms within each group using the Boolean operator "OR" and then linked the two groups with the Boolean operator "AND." During the database search, all collected references were imported into the EndNote software to avoid duplicates from different databases.

3.2 | Inclusion and Exclusion Criteria

To ensure the rigour of this meta-analysis, we established the following inclusion and exclusion criteria:

1. Studies included should examine the impact of ChatGPT on students' academic achievement. Studies focusing on factors affecting students' intention to use ChatGPT were excluded (e.g., Wang et al. 2025).
2. Studies included should use ChatGPT as an intervention tool in the learning process. For example, Ahmed Moneus and Al-Wasy's (2024) study that merely used ChatGPT as a translation tool rather than to improve translation skills was excluded.
3. This meta-analysis included only studies that adopted a (quasi-)experimental design, where the control group did not use ChatGPT (or related tools), while the experimental group did. Studies that combined ChatGPT with other GenAI tools were also excluded from this meta-analysis.
4. To maintain methodological consistency and avoid confounding effects, participants in the control group should use only traditional materials, such as paper and pen, PowerPoint, and basic learning tools required for the course (e.g., computers in programming courses). Studies

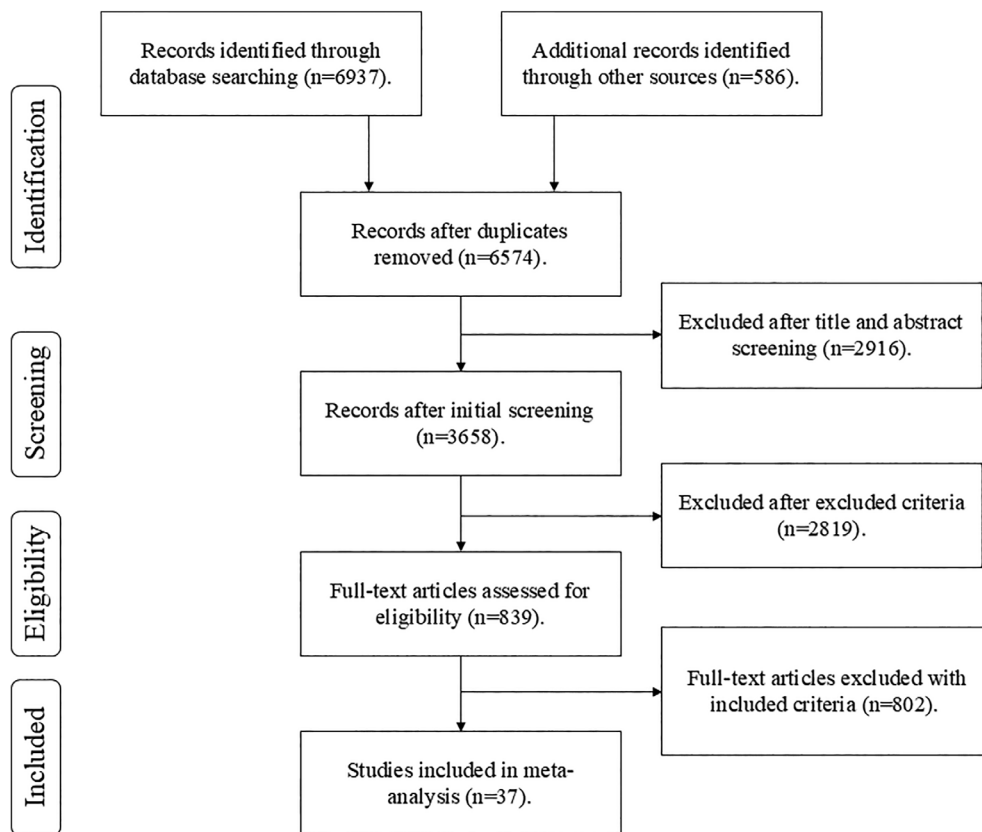


FIGURE 1 | Flow diagram.

where the control group used technologies with intelligent features (e.g., virtual reality) were excluded.

5. The control and experimental groups should employ the same teaching approach. Studies that did not explicitly state that both groups adopted the same teaching approach were excluded (e.g., Hui et al. 2025).
6. Studies included should report quantifiable data on students' academic achievement (means, standard deviations, and sample sizes) to calculate effect sizes. If a delayed test was conducted in the study, data were extracted only from the final test point. If a study reported both subgroup and overall data, the overall data were used.
7. Studies relying solely on qualitative data without quantitative analysis, as well as those using within-subject designs or evaluating results unrelated to student learning, were excluded.
8. This meta-analysis included only journal articles and conference papers published in English after December 2022.

Figure 1 presents the process of literature collection, screening, and inclusion. The initial search yielded 7523 records from multiple databases. After removing 949 duplicate studies, 6574 studies remained for screening. During the abstract and title screening phase, 2916 studies were excluded, leaving 3658 studies for full-text assessment. Based on the inclusion criteria, 839 studies were identified as eligible. Subsequently, 802 studies

were excluded according to the exclusion criteria. Ultimately, 37 studies were included in data coding.

3.3 | Coding

Based on the Cooper's (2016) coding scheme, this study coded the following information from 37 articles: (1) title; (2) publication year; (3) author information; (4) educational level; (5) discipline; (6) intervention duration; (7) sample size; (8) knowledge type; (9) instructional model; (10) role setting; (11) learning approach; (12) generated content. Details of the coding process are provided in the Appendix S1.

As shown in Table 1, the selected moderator variables were categorised based on existing research. Specifically, following the classification criteria for educational level moderators proposed by Deng et al. (2024), we classified them into K-12 and college education. In line with the framework outlined by Sun and Zhou (2024), the moderator variables were further categorised as follows: disciplines (information engineering, natural sciences and social sciences); intervention durations (> 15, 10-15, 5-10, 1-5, and < 1 week); sample sizes (0-20, 21-40, and > 40); learning approaches (collaborative learning or individual learning); generate contents (text or code). Furthermore, knowledge types were categorised into declarative knowledge and procedural knowledge based on the Wang et al.'s (2024) classification strategy for moderator variables. Regarding the use of ChatGPT, we adopted the distinction made by Kong et al. (2023), classifying

TABLE 1 | Moderating variables and their categories.

Moderators	Categories	References
Educational level	K-12, college.	(Deng et al. 2024)
Discipline	Information engineering, natural sciences, social sciences.	(Sun and Zhou 2024)
Intervention duration	<1, 1–5, 5–10, 10–15, > 15 weeks.	
Sample size	0–20, 21–40, > 40.	
Learning approach	Collaborative learning, individual learning	
Generate content	Text, code.	
Knowledge type	Declarative knowledge, procedural knowledge.	(Wang et al. 2024)
Role setting	Yes, no.	(Kong et al. 2023)
Instructional model	Flipped classrooms, traditional classrooms.	(Strelan et al. 2020)

it into role-setting and non-role-setting modes. Finally, the instructional models were categorised as flipped or traditional classrooms according to the criteria established by Strelan et al. (2020). To ensure reliability in the coding process, two authors independently coded all studies with 0.896 consistency.

3.4 | Research Quality Assessment

According to the quality assessment criteria developed by Kmet et al. (2004), the quality of the 37 included studies was assessed. It is a versatile tool designed to evaluate all sections of empirical studies (e.g., study design, method). This tool has proven effective in evaluating the quality of empirical research and has been widely applied in educational research (Deng et al. 2024; Wu and Yu 2024). Each study was evaluated across 14 dimensions, with each item scored as yes (2), partial (1), and no (0). The score for each study was calculated by dividing the total score by 28 points. According to thresholds established in Kmet et al.'s (2004) research, a score of <0.50, 0.50–0.70, 0.71–0.80, and > 0.80 indicates low, adequate, good, and high quality, respectively. In this meta-analysis, the quality scores of the 37 studies ranged from 0.80 to 0.94, indicating that all studies met the quality standards required for meta-analysis. The first and third authors independently evaluated the quality of each study. The inter-rater consistency coefficient was 0.85, indicating a high level of consistency. Any discrepancies in the evaluations were resolved by consulting the corresponding author.

3.5 | Data Analysis

To comprehensively explore the effect of ChatGPT on enhancing students' academic achievement, this study follows the guidelines of the Preferred Reporting Items for Systematic reviews and Meta-Analyses (PRISMA) statement (Page et al. 2021). Comprehensive Meta-Analysis (CMA 3.0) software was employed to conduct effect size calculation, heterogeneity test, moderator analysis, publication bias test, and sensitivity analysis on the coded results. The following section introduces these processes.

3.5.1 | Calculating Effect Size

Effect size is a crucial metric for assessing the strength of correlation between variables or the level of experimental effects (Hedges and Vevea 1998). Given the relatively small sample sizes and the different (quasi-)experimental designs of these studies, Hedges' *g* was selected as the metric for calculating effect sizes due to its ability to correct for small sample bias and reduce bias in standardised mean differences. Unlike Cohen's *d*, which may overestimate effect size in small or unequal samples, Hedges' *g* offers a more conservative and unbiased estimate. This makes it suitable for educational studies with different designs and sample sizes. According to Güler et al.'s (2022) study, an effect size between 0.15–0.40, 0.40–0.75, and > 0.75 is interpreted as low, medium, and large, respectively.

Additionally, to obtain a single effect size from multiple comparisons within a single study, a combined effect size was calculated using “the study as the unit of analysis” and “subgroups within the study” options in CMA 3.0 (Çalik et al. 2023, 2024). In other words, this study aims to minimise the number of individual studies to address the inflating effects caused by multiple experimental outcomes (Rahman and Lewis 2020). It is not recommended to treat different results from the same participants as independent outcomes, as this may result in a misestimation of the overall effect.

3.5.2 | Heterogeneity Test and Moderator Analysis

To choose between the random-effects model and the fixed-effects model, the researchers considered how the characteristics of the included papers matched the prerequisites of each model. The selection of a meta-analysis model should be determined by the assumptions regarding the distribution of true effect sizes across studies and the intended scope of inference (Borenstein 2019). The fixed-effects model assumes that all studies share a common true effect size and that observed differences among studies are due to sampling error. According to Hedges and Vevea (1998), the fixed-effects model allows for inferences to be made about the relevant parameters within the analysed studies. It is suitable when the studies are functionally

TABLE 2 | Overall effect size.

Model	Effect size	95% CI	Q	Homogeneity test			Tau-squared		
				df	p	I ²	Tau	Tau ²	SE
Fixed	0.398	(0.356, 0.439)	530.195	36	0.000	93.210%	0.505	0.255	0.124
Random	0.577	(0.395, 0.759)							

Abbreviation: CI, confidence interval.

identical, estimating the effect size for a single population. In contrast, the random-effects model assumes that true effect sizes may vary across studies due to the influence of moderators (e.g., educational level). When studies are functionally different, the random-effects model can estimate and generalise the effect size to a broader population.

Building on these theoretical foundations, the researchers also examined heterogeneity by calculating the *Q*-value and *I*² statistic. The heterogeneity criteria established by Higgins et al. (2003) were used to guide model selection. A significant *Q*-test indicates the presence of heterogeneity in the sample data, where the variability in effect sizes exceeds what would be expected due to sampling errors. The *I*² statistic reflects the proportion of total variance in effect sizes attributable to heterogeneity, with *I*² values of 25%, 50%, and 75% corresponding to low, medium, and high levels of heterogeneity, respectively.

Overall, significant heterogeneity indicates that the impact of ChatGPT on students' academic achievement may be influenced by potential moderators. Therefore, further *Q*-tests across moderator variables are necessary to assess whether significant differences in effect sizes exist.

3.5.3 | Publication Bias and Sensitivity Analyses

Publication bias refers to the tendency for studies with statistically significant results or large effect sizes to be more likely to be published than those with non-significant findings or smaller effect sizes (Borenstein et al. 2021). Based on the distribution of effect size, a funnel plot was used to evaluate publication bias. However, this method relies on qualitative visual inspection of the plot, resulting in a degree of subjectivity in the conclusion. Therefore, to achieve symmetry in the funnel plot, the trim-and-fill method was employed to estimate the number of missing studies.

To further quantify the effect of publication bias, we applied Orwin's fail-safe *N* to evaluate the number of missing studies required to reduce the effect size to a negligible level ($g < 0.01$). Moreover, the Classic fail-safe *N* test was employed in the study to further assess whether existing publication bias could undermine the overall conclusions (Littell et al. 2008). This method estimates the minimum number of missing studies needed to nullify the observed effect size. Following Rosenthal's (1995) criterion, the threshold is set at $5n + 10$, where *n* represents the number of included studies. A fail-safe *N* larger than this threshold suggests that the effect of unpublished publications on the overall effect is likely minimal.

Finally, sensitivity analyses were conducted to identify outliers that influence the overall effect. A single-study removal method was employed to assess the effect of studies with extreme negative or positive effects. Furthermore, studies with small samples that caused large heterogeneity were excluded to improve the stability of the results.

4 | Results

4.1 | Descriptive Information

Of the 37 articles included in the study, the experimental group consisted of 1962 participants, while the control group included 2147 participants. The range of effect sizes was (−1.856, 4.203), with 7 negative outcomes and 30 positive ones.

4.2 | Heterogeneity Test

Table 2 presents the results of the heterogeneity test. The *Q*-value of the overall effect was 530.195 ($df = 36$, $p < 0.001$), indicating heterogeneity among the samples. The *I*² value was 93.210%, which was greater than 75%, showing high heterogeneity. Therefore, this study adopted the random-effects model to estimate effect size and employed moderator analysis to explore the sources of heterogeneity.

4.3 | Overall Effectiveness

As shown in the forest plot (Figure 2). Under the random-effects model, the overall effect size of 37 studies (37 effect sizes) was 0.577, indicating a moderately positive effect of ChatGPT on students' academic achievement. The two-tailed *Z* test was significant ($Z = 6.224$, 95% CI [0.395, 0.759], $p < 0.001$), indicating that ChatGPT can significantly enhance students' academic achievement.

4.4 | Moderator Analysis

Table 3 shows the results of the moderator analysis. The study used the random-effects model to examine whether the effect size across the analysed studies varied based on moderator variables, aiming to identify sources of heterogeneity.

For the effect size of each educational level, using ChatGPT to teach K–12 students ($g = 0.606$, 95% CI [0.087, 1.125], $p < 0.05$) and college students ($g = 0.570$, 95% CI [0.377, 0.763],

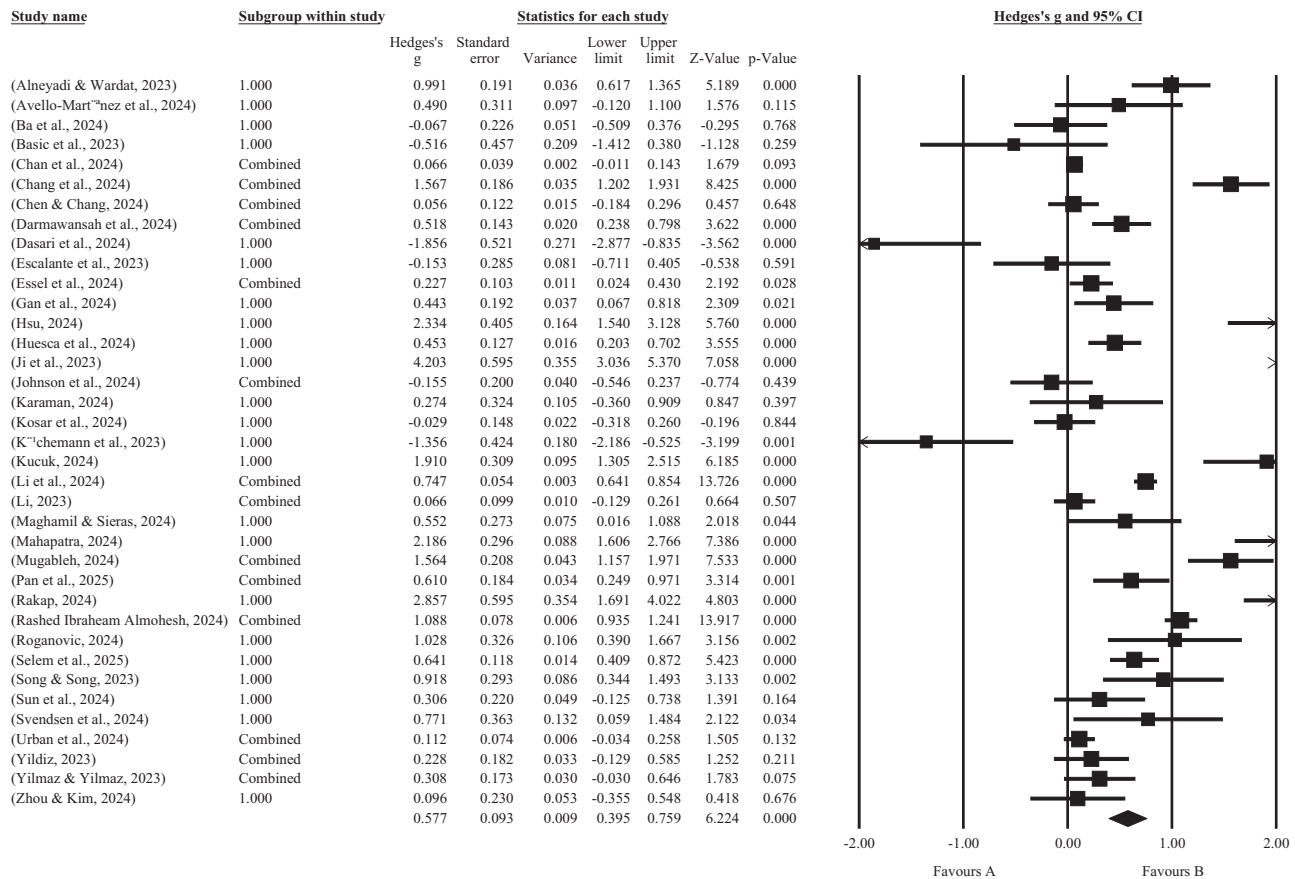


FIGURE 2 | Forest plot.

$p < 0.001$) had a moderately positive effect on students' academic achievement. However, the intergroup effect test ($Q_B = 0.016$, $p > 0.05$) showed no significant differences in the impact of ChatGPT on students' academic achievement across different educational levels.

For the effect size of each discipline, social sciences had a moderate effect ($g = 0.696$, 95% CI [0.469, 0.992], $p < 0.001$), followed by a moderate effect of natural sciences ($g = 0.561$, 95% CI [-0.041, 1.163], $p > 0.05$), and a small effect size of information engineering ($g = 0.187$, 95% CI [-0.048, 0.422], $p > 0.05$). However, the effect of natural sciences and information engineering did not reach statistical significance. Overall, the intergroup effect was significant ($Q_B = 9.442$, $p < 0.01$), indicating a significant difference in the impact of ChatGPT on students' academic achievement across different disciplines.

For the effect size of each intervention duration, the effect size was large when studies lasted 5–10 weeks ($g = 0.807$, 95% CI [0.427, 1.186], $p < 0.001$) and > 15 weeks ($g = 0.761$, 95% CI [-0.800, 2.322], $p > 0.05$), followed by 1–5 weeks ($g = 0.700$, 95% CI [-0.439, 1.839], $p > 0.05$), 10–15 weeks ($g = 0.564$, 95% CI [0.330, 0.798], $p < 0.001$), < 1 week ($g = 0.484$, 95% CI [0.139, 0.830], $p < 0.01$), and all of which achieved moderate effects. However, the > 15 weeks and 1–5 weeks durations did not reach statistical significance. Overall, the intergroup effect was significant ($Q_B = 23.324$, $p < 0.001$), indicating a significant difference in the impact of ChatGPT on students' academic achievement across different intervention durations.

For the effect size of each sample size, 21–40 participants had a large effect ($g = 1.214$, 95% CI [0.066, 2.362], $p < 0.05$), followed by a moderate effect of > 40 sample size ($g = 0.576$, 95% CI [0.395, 0.757], $p < 0.001$). Moreover, the effect of 0–20 sample sizes ($g = -1.162$, 95% CI [-2.475, 0.150], $p > 0.05$) was large but not statistically significant. The intergroup effect was significant ($Q_B = 7.895$, $p < 0.05$), suggesting that the improvement in students' academic achievement varied significantly across different sample sizes.

For the effect size of each knowledge type, the large effect of declarative knowledge ($g = 1.411$, 95% CI [0.840, 1.982], $p < 0.001$) was greater than the moderate effect of procedural knowledge ($g = 0.487$, 95% CI [0.304, 0.671], $p < 0.001$). Furthermore, the intergroup effect was significant ($Q_B = 9.103$, $p < 0.01$), indicating that students' academic achievement varied significantly when using ChatGPT to learn different types of knowledge.

For the effect size of each instructional model, the moderate effect of traditional classrooms ($g = 0.627$, 95% CI [0.420, 0.835], $p < 0.001$) was greater than the small effect of the flipped classroom ($g = 0.246$, 95% CI [0.078, 0.414], $p < 0.01$). The intergroup effect was significant ($Q_B = 7.848$, $p < 0.01$), indicating that there was a significant difference in the impact of ChatGPT on students' academic achievement across different instructional models.

For the effect size of each role setting, the moderate effect of ChatGPT with a role ($g = 0.540$, 95% CI [0.221, 0.859], $p < 0.01$)

TABLE 3 | Moderator analyses.

Moderator	<i>k</i>	<i>g</i>	<i>z</i>	95% CI	<i>Q_B</i>
Educational levels					
K–12	5	0.606	2.290*	(0.087, 1.125)	0.016
College	32	0.570	5.798***	(0.377, 0.763)	
Disciplines					
Natural science	10	0.561	1.828	(−0.041, 1.163)	9.442**
Social science	22	0.696	6.017***	(0.469, 0.992)	
Information engineering	5	0.187	1.560	(−0.048, 0.422)	
Intervention durations					
< 1 week	13	0.484	2.752**	(0.139, 0.830)	23.324***
1–5 weeks	3	0.700	1.205	(−0.439, 1.839)	
5–10 weeks	11	0.807	4.163***	(0.427, 1.186)	
10–15 weeks	7	0.564	4.721***	(0.330, 0.798)	
> 15 weeks	2	0.761	0.955	(−0.800, 2.322)	
No report	1	−1.856	−3.562***	(−2.877, −0.835)	
Sample sizes					
0–20	2	−1.162	−1.736	(−2.475, 0.150)	7.895*
21–40	7	1.214	2.072*	(0.066, 2.362)	
> 40	28	0.576	6.246***	(0.395, 0.757)	
Knowledge types					
Declarative knowledge	4	1.411	4.842***	(0.840, 1.982)	9.103**
Procedural knowledge	33	0.487	5.199***	(0.304, 0.671)	
Instructional models					
Traditional classroom	33	0.627	5.924***	(0.420, 0.835)	7.848**
Flipped classroom	4	0.246	2.868**	(0.078, 0.414)	
Role setting					
Yes	12	0.540	3.322**	(0.221, 0.859)	0.097
No	25	0.604	4.869***	(0.361, 0.847)	
Learning approach					
Collaborative learning	4	0.868	2.246*	(0.111, 1.625)	0.594
Individual learning	33	0.561	5.743***	(0.369, 0.752)	
Generate contents					
Text	33	0.624	6.080***	(0.423, 0.825)	5.470*
Code	4	0.260	2.222*	(0.031, 0.489)	

Abbreviations: *k* = number of effect size; *g* = effect size; *Q_B* = intergroup homogeneity.**p* < 0.05.***p* < 0.01.****p* < 0.001.

was smaller than the moderate effect of ChatGPT without a role ($g = 0.604$, 95% CI [0.361, 0.847], $p < 0.001$). The intergroup effect was not significant ($Q_B = 0.097$, $p > 0.05$), indicating that students' academic achievement did not vary significantly between using ChatGPT with or without role setting.

For the effect size of each learning approach, the large effect of using ChatGPT in collaborative learning ($g = 0.868$, 95% CI [0.111, 1.625], $p < 0.05$) was greater than the moderate effect observed in individual learning ($g = 0.561$, 95% CI [0.369, 0.752], $p < 0.001$). The intergroup effect was not significant ($Q_B = 0.594$,

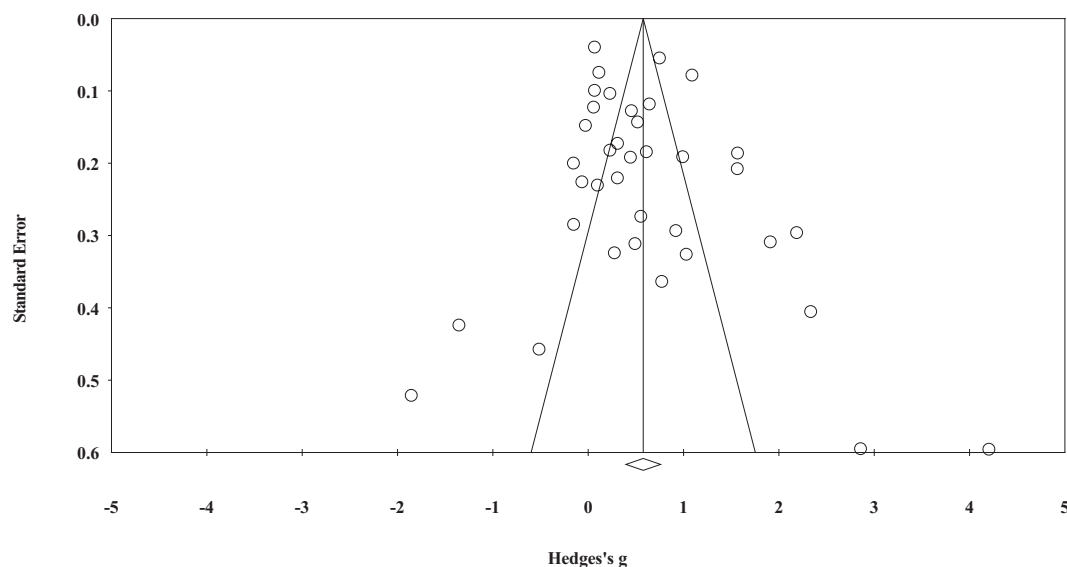


FIGURE 3 | Funnel plot.

TABLE 4 | Results of Orwin's fail-safe N .

Hedges' g in observed studies (random effect)	0.40
The criterion for a 'trivial' Hedges' g	0.01
Mean Hedges' g in missing studies	0.00
Number missing studies needed to bring Hedge's g under 0.01	1435.00

$p > 0.05$), indicating no significant difference in the impact of ChatGPT on students' academic achievement across different learning approaches.

For the effect size of each generating content, the moderate effect of text ($g = 0.624$, 95% CI [0.423, 0.825], $p < 0.001$) was greater than the small effect of code ($g = 0.260$, 95% CI [0.031, 0.489], $p < 0.01$). The intergroup effect was significant ($Q_B = 5.470$, $p < 0.05$), indicating a significant difference in students' academic achievement across different contents generated by ChatGPT.

4.5 | Publication Bias Test and Sensitivity Analysis

The funnel plot (Figure 3) indicated that the effect sizes from most studies were symmetrically distributed around the mean effect size. The trim-and-fill method also revealed that four additional studies would be added to the right side of this funnel plot to achieve symmetry. These suggested a low likelihood of publication bias.

Moreover, Orwin's fail-safe N test (see Table 4) indicated that, at a 0.01 significance level, 1435 unpublished studies would be needed to significantly alter the total effect size. Classic fail-safe N test (see Table 5) also indicated that 3270 unpublished studies would be required to reduce the overall effect size, far exceeding the threshold of 195 studies. Additionally, Egger's test ($p = 0.11 > 0.05$) and Kendall's Tau rank test

TABLE 5 | Results of Classic fail-safe N .

z -value for observed studies	18.53
p -value for observed studies	0.00
Alpha	0.05
Tails	2.00
z for alpha	1.96
Number of observed studies	37.00
Number of missing studies that would bring p -value to $> \alpha$	3270.00

($p = 0.29 > 0.05$) did not reach statistical significance. Based on these tests, we found no evidence of publication bias, suggesting that the validity of the data included in this meta-analysis was high.

Finally, the result of sensitivity analysis showed that excluding a certain study did not affect the meta-analysis outcomes, with all estimates falling within the range of 95% CI (0.395, 0.759), indicating that the model was robust.

5 | Discussion

This study employs meta-analysis to examine 37 studies published since 2022 on the impact of ChatGPT on students' academic achievement. Additionally, it examines the differential effects of moderator variables.

5.1 | ChatGPT Can Enhance Students' Academic Achievement

The study found that ChatGPT has a moderately positive impact on students' academic achievement, with an overall effect size of 0.577. The overall effect indicates that ChatGPT is effective

in enhancing students' academic achievement. This finding is consistent not only with the meta-analysis findings of Deng et al. (2024) and Heung and Chiu (2025) but also with the results of most empirical studies (Mahapatra 2024; Yıldız 2023). Specifically, ChatGPT offers a personalised learning experience and timely feedback, encouraging students to engage more deeply in learning activities (Maghamil and Sieras 2024). Furthermore, the interactivity of ChatGPT aligns with constructivist learning theory, which emphasises that learning is not merely a process of knowledge transfer but rather a process of knowledge construction through deep interaction (Debets et al. 2025). Through the interaction between humans and ChatGPT, the learning process is driven, enabling the collaborative creation of knowledge. Interestingly, this study compares the current findings with the results of another meta-analysis that focused on the effects of AI-based assessment tools on students' learning achievement (Chen et al. 2025). The results indicate that the effect of ChatGPT on students' academic achievement is significantly larger than that of other AI tools ($g=0.390$). Our explanation is that ChatGPT overcomes the limitations of shallow interaction, low efficiency, and preset feedback commonly found in traditional human-machine learning systems. Its powerful capabilities in language translation and contextual understanding meet students' diverse needs (Chan et al. 2024; Roganović 2024). In this way, ChatGPT supports learners in collaborative knowledge creation through brainstorming in human-machine interactions.

5.2 | The Moderator Analysis

This study investigates nine moderator variables to identify factors that may explain differences in effect sizes between studies. The results found that, in addition to educational level, role-setting, and learning approach, differences in discipline, intervention duration, sample size, knowledge type, instructional model, and generated content were key factors influencing the effect of ChatGPT on students' academic achievement.

From the perspective of educational level, although ChatGPT has a positive effect on academic achievement across different levels, the moderating effect of educational level on ChatGPT's impact is not significant. This does not support Zhu et al.'s (2025) findings that GenAI is more effective in higher education than in secondary education. This may be because the development of more intuitive interfaces challenges preconceptions regarding age-related efficacy (Laun and Wolff 2025). GenAI tools such as ChatGPT are increasingly designed to be user-friendly and accessible, enabling learners of all ages to use them effectively. Even younger students can benefit from ChatGPT-supported learning when guided by appropriate instructional support. This growing technological accessibility likely contributes to the consistency of learning outcomes across educational levels, as supported by Deng et al.'s (2024) findings. Furthermore, given the limited number of available studies, we combined elementary and secondary education into a single K–12 category. However, this categorization may reduce the differences between primary and secondary students, limiting the intergroup effects of the moderator variable. Therefore, this finding should be interpreted with caution. While educational level is not a significant moderator in this study, the result reflects the technological universality of ChatGPT and limitations of the current study

classification. It underscores the need for more representative studies to understand how ChatGPT interacts with students at different educational levels.

From the perspective of discipline, using ChatGPT significantly improves students' academic achievement in social sciences, while no significant improvement is observed in natural sciences and information engineering, with notable intergroup differences. This is consistent with previous studies, for instance, Kosar et al. (2024) found that students' academic achievement on programming assignments was not influenced by using ChatGPT. Debugging algorithms requires students to continually modify and optimise their work to master the skills involved. While ChatGPT can serve as a useful tool to help students solve certain algorithmic problems, its usefulness is limited when it comes to tasks requiring the precise application of technical skills (Johnson et al. 2024). Furthermore, the application of ChatGPT in the natural sciences is also limited. The reason may be that ChatGPT cannot directly perform complex formula derivations, which limits its use in such disciplines and may negatively impact learning outcomes (Küchemann et al. 2023). In contrast, the social sciences have a strong social and humanistic nature, which enables ChatGPT to simulate real social contexts (Mugableh 2024). It interacts with learners through human-like conversation, offering immersive learning experiences that effectively facilitate knowledge understanding.

From the perspective of intervention duration, the effect is greatest at 5–10 weeks, with a decline observed when the duration is either shorter or longer. This suggests that a moderate intervention duration is more conducive to learning outcomes, which contrasts with previous research (Wang and Fan 2025). When students are first introduced to ChatGPT, they often show strong curiosity and interest, which can positively influence short-term achievement (Wu and Yu 2024). However, during very short interventions (e.g., <1 week), their limited proficiency in using ChatGPT and insufficient critical thinking may limit learning effectiveness (Avello-Martínez et al. 2024). As the intervention duration increases, students become more skilled and strategic in using ChatGPT, which may improve their academic achievement. Nonetheless, when the intervention duration exceeds a certain threshold (i.e., >10 weeks), the learning benefits seem to decline. One possible explanation is that long-term use of ChatGPT may reduce students' intrinsic motivation, leading to over-reliance on AI-generated outputs. Moreover, prolonged use may cause problems such as information overload and reduced attention, further reducing learning efficiency (Gan et al. 2024). These findings suggest that while sustained interaction with ChatGPT can support knowledge acquisition, its educational value may diminish without metacognitive monitoring or self-regulation strategies (Urban et al. 2024).

From the perspective of sample size, ChatGPT has a significant positive effect on students' academic achievement when the sample size is between 21 and 40, with notable intergroup differences. This finding is consistent with Bucea-Manea-Țoniș et al.'s (2022) study. Specifically, when the sample size is between 21 and 40, the group achieves a good balance of diversity and interactivity, allowing teachers to effectively monitor each student's learning progress. The combination of ChatGPT assistance and teacher guidance enables students to receive

system-generated feedback and real-time instruction in class. A moderate-sized group ensures that students receive the instructor's attention within a reasonable timeframe while avoiding overburdening teaching resources (Ji et al. 2023). However, when the number of students exceeds 40, the available learning resources in the classroom become limited, and some students may not receive sufficient learning support (Hobert et al. 2023). Large-scale classrooms may lead to insufficient learning depth and reduce the efficiency of students' interaction with ChatGPT. Notably, when the sample size is between 0 and 20, the effect of ChatGPT on students' academic achievement is not statistically significant. One possible explanation is that ChatGPT's potential has not been fully realised in very small sample, or that other unconsidered factors may affect its effectiveness (Dasari et al. 2024).

From the perspective of knowledge types, ChatGPT has a greater positive impact on declarative knowledge than on procedural knowledge, with notable intergroup differences. Specifically, declarative knowledge includes facts and concepts, which ChatGPT can break down and present in a structured manner, making them easier for learners to grasp (Hsu 2024). In contrast, procedural knowledge involves not only specific operational skills but also requires practical feedback. While ChatGPT can assist learners by explaining procedures or providing guidance for skill practice, it cannot fully replace the experiential learning gained from hands-on practice (Karaman 2024). Furthermore, with the aid of ChatGPT, academic achievement in declarative knowledge is generally more predictable, whereas achievement in procedural knowledge is more dynamic, with greater uncertainty (Selem et al. 2025).

From the perspective of instructional models, the combination of traditional classrooms with ChatGPT is more effective than the flipped classroom. This result contrasts with previous studies; for example, Strelan et al. (2020) suggested that the flipped classroom provides learners with a flexible human-computer interaction space, effectively fostering diverse cognitive perspectives. However, not all students possess high self-management ability. For those lacking time-planning skills, the flipped classroom may lead to delayed learning progress (Cheng et al. 2019). If students in both traditional and flipped classrooms are sufficiently motivated to seek help proactively, the effect of ChatGPT may be similar. Furthermore, for students unfamiliar with ChatGPT, implementing the flipped classroom may be more challenging, particularly in low-resource environments where students do not fully use ChatGPT to enhance their learning experience (Yilmaz and Yilmaz 2023). In contrast, students in traditional classrooms benefit from instructors' immediate feedback and ChatGPT's personalised support, which can significantly improve learning outcomes.

From the perspective of role setting, ChatGPT has no significant effect on students' academic achievement with or without specific roles. This is inconsistent with existing study (Kong et al. 2023). For example, Chang et al. (2024) found that role setting facilitates the segmentation of mixed roles, helping ChatGPT exhibit more nuanced and coherent interactions. This may be attributed to the nature of the learning task. Role setting may be particularly effective in complex problem-solving tasks where specific guidance is crucial (Rakap 2024). However,

if a task requires less interaction or more self-directed learning, role setting may not have the same positive effect (Escalante et al. 2023). Furthermore, if an ineffective implementation of role settings is used, the system may not exhibit its effectiveness, thereby limiting its potential impact on students' academic achievement. Further exploration of these factors in future research will be beneficial to better understand the conditions under which role-setting has the most significant effect on students' academic achievement.

From the perspective of learning approach, ChatGPT significantly enhances students' academic achievement under two conditions. This finding contrasts with Slavin's (2004) study that collaborative learning is more conducive to enhancing students' academic achievement. One possible explanation is that students interact not only with ChatGPT but also with their peers in a collaborative environment. Frequent chat interactions may distract students, thereby reducing its overall impact on academic achievement (Sánchez-Ruiz et al. 2023). In contrast, students can work at their own pace in individual learning environments, receiving personalised guidance (Pan et al. 2025). However, no significant difference was found between the groups, indicating that individual learning also has its limitations. Individual learning often lacks the depth of peer dialogue, which may lead to shallow exploration of the content. The interaction between the learner and ChatGPT cannot fully replace human-human communication. This limitation may restrict ChatGPT's benefits in an individual learning environment, resulting in no significant difference in academic achievement compared to collaborative learning.

From the perspective of generated contents, both text and code positively influenced academic achievement, with a significant difference observed between the groups. This finding is consistent with the study by Sun and Zhou (2024), which reported a higher effect of GenAI in text generation. The generated texts may be more easily integrated into the learning process due to their ease of assessment and logical coherence, thus having a greater effect on academic achievement (Darmawansah et al. 2025). In contrast, the generated code, which may be more difficult to assess and technologically immature, has a more limited effect on academic achievement (Kosar et al. 2024). Specifically, students are required to repeatedly debug and modify the code to find potential problems, a process that often involves psychological burden and frustration. As a result, while code generation enhances students' programming skills, its impact on academic achievement may require more time and practice to fully realise its benefits (Yilmaz and Yilmaz 2023).

6 | Implications

This study has important theoretical and practical implications for education. Theoretically, the study expands the existing research on ChatGPT and academic achievement by offering quantitative evidence that ChatGPT is an effective tool for improving students' academic achievement. This study also offers a more comprehensive and thorough investigation of moderator variables. It examines the impact of ChatGPT on students' academic achievement across nine different moderators. It expands the scope compared to previous meta-analyses

(Heung and Chiu 2025). Unlike the meta-analysis of Deng et al. (2024) and Wang and Fan (2025), this study further examines the impact of ChatGPT on the learning process while controlling other confounding factors (e.g., learning methods, teaching approaches) that may influence its effectiveness. This study provides a theoretical foundation for the future development of national GenAI education policies and the use of ChatGPT by educators, while further enriching the research on ChatGPT's application in education. Practically, ChatGPT-assisted learning is not suitable for all learning environments and cannot be used arbitrarily. Future work should focus on the following nine points:

1. Teachers should develop targeted teaching strategies that cater to the educational needs of students at different levels. Students should also receive the necessary technical training before the lessons. Future research should address the current gap in the literature by conducting more empirical studies at the K–12 educational level. Developers should simplify the user interface to enhance usability for younger students with limited technical skills.
2. For social sciences, teachers can use ChatGPT to present knowledge in diverse scenarios. Teachers should also encourage students to use ChatGPT to validate algorithmic ideas or debug code. For natural sciences, teachers can use ChatGPT to simplify abstract content, enhancing learners' understanding of derivation processes. Future work should explore the integration of ChatGPT with virtual tools, which can compensate for ChatGPT's limitations in hands-on experimentation.
3. In the initial stage of ChatGPT-assisted learning, teachers should prevent learners from becoming overly absorbed in the novelty effect. In the later stages, teachers should strengthen the incentives for cognitive participation, alleviating the cognitive fatigue caused by the diminishing novelty effect. Researchers should develop adaptive models that balance novelty and over-reliance to maintain learner interest and prevent burnout.
4. Teachers should work closely with each student in 0–20 samples, offering personalised instruction. For 21–40 samples, teachers should combine real-time guidance with ChatGPT's feedback, designing hierarchical teaching tasks. For > 40 sample size, teachers should use the flipped classroom to maximise the use of limited classroom resources. Future research should employ qualitative interviews or classroom observations to explore unknown variables in small-sample contexts (e.g., low learning motivation or limited ChatGPT usage).
5. It is recommended to use ChatGPT for learning declarative knowledge in brainstorming-based learning environments; for procedural knowledge, teachers should focus on using ChatGPT to provide simulated operations. Teachers should also analyse the dialogue process in ChatGPT, aiming to establish an evaluation system centred on the learning process. Future research should explore hybrid learning models that integrate ChatGPT with diverse feedback mechanisms to enhance the acquisition of different types of knowledge.
6. To implement a flexible blended instructional model, teachers should design a triadic intelligent interaction between students and ChatGPT. Teachers should also avoid students becoming overly dependent on ChatGPT and ensure that those with lower technical proficiency can keep up with the learning process. A promising direction for future work is to explore a hybrid instructional model that combines flipped classroom elements with traditional classroom settings and ChatGPT support. This approach retains the supervisory advantages of traditional teaching while incorporating the flexibility of the flipped classroom, potentially identifying an optimal combination to enhance students' academic achievement.
7. Teachers should train students to use different levels of prompts to ensure the accuracy of output. Developers should design an easily integrable role-setting module that enables educators to swiftly configure ChatGPT's attributes (e.g., knowledge scope, feedback style) to align with diverse instructional goals and student needs.
8. Educators should consider the alignment between task types and ChatGPT. Self-reflection tasks are suited for individual learning, while collaborative learning is more effective for creative problem-solving tasks. In collaborative learning, teachers should use time as a guiding framework to avoid endless debates. In individual learning, developers can design AI-guided modules that simulate peer feedback to enhance interactivity. In collaborative learning, a simple interaction interface should be designed to avoid cognitive overload.
9. Teachers should encourage students to use ChatGPT in writing to produce well-structured texts. Meanwhile, teachers should provide suggestions for revisions, assisting students in refining their work. For programming tasks, students can use ChatGPT to generate code, accompanied by requesting optimisation suggestions. Developers should design a specialised version of ChatGPT tailored for code generation and writing tasks, which can integrate task-specific capabilities (e.g., to enhance logical reasoning for programming).

7 | Limitations

This study has several limitations. First, some moderators have small sample sizes, which restrict the study of interactions between moderator variables. Future work should expand sample sizes to further investigate the interaction effect of moderators, such as those between sample size and learning approach. Second, due to the researchers' language limitations, non-English studies were excluded. Future work that incorporates studies in other languages can provide a more comprehensive understanding of the impact of ChatGPT on students from different language backgrounds. Third, this study focuses solely on the impact of ChatGPT, without considering other GenAI tools. Although ChatGPT is widely applied, other tools also provide unique functionalities that meet various instructional needs. Future work should investigate the impact of different tools to gain a more nuanced understanding of how different tools

influence learning in different educational conditions. Finally, the present study included only (quasi-)experimental studies with available data for meta-analysis, excluding related qualitative research. Future work should combine qualitative analysis and meta-analysis to conduct a more comprehensive analysis.

8 | Conclusion

This study uses a meta-analysis method to examine the effect of ChatGPT on students' academic achievement. By analysing 37 studies and calculating 37 effect sizes, the results reveal a moderately positive impact of ChatGPT on students' academic achievement. Further analysis of moderator variables indicates that ChatGPT has no significant impact on students' academic achievement in different educational levels, role setting, and learning approaches, but significant differences are observed in different disciplines, intervention durations, sample size, knowledge types, instructional models, and generated contents. In conclusion, the findings suggest that the positive effect of ChatGPT on students' academic achievement and its impact varies under different learning contexts and application conditions. Based on these findings, governments can implement policies to promote the integration of ChatGPT into education, encouraging learners to apply ChatGPT appropriately.

Author Contributions

Zhiwei Liu: conceptualization, methodology, software, data curation, investigation, validation, formal analysis, visualization, writing – original draft. **Haode Zuo:** conceptualization, methodology, supervision, writing – review and editing, writing – original draft, resources, project administration, visualization. **Yongjing Lu:** formal analysis, writing – review and editing.

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Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The data is available upon reasonable request.

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Supporting Information

Additional supporting information can be found online in the Supporting Information section.